

PREDICATE LOGIC

CSE 2213 – Discrete Mathematics

LOGICAL EQUIVALENCES INVOLVING QUANTIFIERS

- Every CS student learns programming AND discrete mathematics.
- $\forall x (P(x) \wedge Q(x))$
- $P(x)$: *x learns programming*
- $Q(x)$: *x learns DM*
- Every CS student learns programming AND every CS student learns discrete mathematics.
- $(\forall x P(x)) \wedge (\forall x Q(x))$

LOGICAL EQUIVALENCES INVOLVING QUANTIFIERS

- Some CS student excels in hardware OR software.
- $\exists x(P(x) \vee Q(x))$
- Some CS student excels in hardware OR some CS student excels in software.
- $(\exists x P(x)) \vee (\exists x Q(x))$

NEGATING QUANTIFIED EXPRESSIONS

De Morgan's Laws for Quantifiers

- $\neg(\forall x P(x)) \equiv \exists x \neg P(x)$
- $\neg(\exists x P(x)) \equiv \forall x \neg P(x)$

It's NOT that every CS student has taken calculus.

$\equiv$ Some CS student has not taken calculus.

It's NOT that some CS student has taken calculus.

$\equiv$ Every CS student has not taken calculus.

ROSEN, PAGE 41, EXAMPLE 21

- What are the negations of these statements?
 - $\forall x(x^2 > x)$
 - $\exists x(x^2 = 2)$
 - Domain: All positive integers

ROSEN, PAGE 41, EXAMPLE 20

- What are the negations of the statements "There is an honest politician" and "All Americans eat cheeseburgers"?

TRANSLATING FROM ENGLISH TO LOGICAL EXPRESSIONS

ROSEN, PAGE 42, EXAMPLE 24, PART I

- Some student in this class has visited Mexico
- There is a student in this class who visited Mexico
- There is a student x in this class such that x visited Mexico
- There is a person x such that x is a student in this class and x visited Mexico (Domain: all people)
- $\exists x (S(x) \wedge P(x))$
 - Here, $S(x) \equiv x$ is a student in this class,
and $P(x) \equiv x$ visited Mexico

ROSEN, PAGE 42, EXAMPLE 24, PART I

- Express the statement "Every student in this class has studied calculus" using predicates and quantifiers. First, the domain consists of students in this class and second, the domain consists of all people
- $C(x)$, which is the statement " x has studied calculus." if the domain for x consists of the students in the class, we can translate our statement as $\forall x C(x)$.
- "For every person x , if person x is a student in this class then x has studied calculus."
- $\forall x (S(x) \rightarrow C(x))$.

EXAMPLE

ROSEN, PAGE 44, EXAMPLE 26

- Express the following statements using the given predicates and quantifiers:
 - All lions are fierce.
 - Some lions do not drink coffee.
 - Some fierce creatures do not drink coffee.
- Given predicates:
 - $L(x) \equiv x$ is a lion
 - $F(x) \equiv x$ is a fierce creature
 - $C(x) \equiv x$ drinks coffee
- The domain consists of **ALL CREATURES**

EXAMPLE

ROSEN, PAGE 45, EXAMPLE 27

- Express the following statements using the given predicates and quantifiers:
 - All hummingbirds are richly colored.
 - No large birds live on honey.
 - Birds that do not live on honey are dull in color.
 - Hummingbirds are small.
- Given predicates:
 - $P(x) \equiv x$ is a hummingbird
 - $Q(x) \equiv x$ is large
 - $R(x) \equiv x$ lives on honey
 - $S(x) \equiv x$ is richly colored

NESTED QUANTIFIER

- For all values of x and y , $x + y = y + x$
- How to represent this using quantifiers?

NESTED QUANTIFIERS

- Quantifier used within the scope of another quantifier
- The example from previous slide: $\forall x \forall y (x + y = y + x)$
- We can put any quantifier within the scope of any quantifier
- Quantifiers are read left-to-right

ANOTHER EXAMPLE:

- For every real number x , there is a corresponding real number y such that $x + y = 0$.
- How do we represent it using quantifiers?
- $\forall x \exists y (x + y = 0)$

ORDER OF QUANTIFIERS

- Commutative law: $\forall x \forall y (x + y = y + x)$
- Can we rewrite this as $\forall y \forall x (x + y = y + x)$??
- What does it mean?
- Is the original meaning changed?

ORDER OF QUANTIFIERS

- Some random observation: $\exists x \exists y (xy = 6)$
- Can we rewrite this as $\exists y \exists x (xy = 6)$??
- What does it mean?
- Is the original meaning changed?

ORDER OF QUANTIFIERS

- $\forall x \exists y (x + y = 0)$
- Can we write it as $\exists y \forall x (x + y = 0)$??
- What does it mean?
- Is the original meaning changed?

ORDER OF QUANTIFIERS

- If only universal or only existential quantifiers are nested, their order can be changed
- If both quantifiers are nested, their order cannot be changed, otherwise the meaning will be changed

QUICK EXERCISE

Find out the truth values of the following:

- $\forall x \exists y (x + y = 0)$
- $\exists y \forall x (x + y = 0)$